NATURAL LANGUAGE PROCESSING

Introduction, Core Techniques, and Semantic Search

A sample document for testing PDF upload, text extraction, chunking, embeddings, semantic retrieval, and question answering in DocuMind AI.

Topics covered	NLP basics, text preprocessing, embeddings, semantic search, RAG, evaluation, applications
Suggested test	Upload this PDF, ask the sample questions near the end, and check whether the app cites the correct page.
Document structure	Four short sections plus a question bank and glossary.

Prepared as a practical test document

1. What Is Natural Language Processing?

Natural Language Processing (NLP) is a field of artificial intelligence that helps computers work with human language. It combines ideas from computer science, linguistics, statistics, and machine learning. NLP systems can process written text or spoken language to identify patterns, extract information, classify content, and generate responses.

Common NLP tasks

Task	Description	Example
Text classification	Assign a category to text.	Mark an email as spam or not spam.
Sentiment analysis	Estimate the opinion or emotion expressed.	Classify a review as positive or negative.
Named entity recognition	Identify names, places, dates, and other entities.	Find a person's name in a news article.
Machine translation	Translate text between languages.	Translate English text into Marathi.
Summarization	Create a shorter version that preserves key information.	Summarize a long report.
Question answering	Answer a question using available information.	Ask a question about an uploaded PDF.

Why language is difficult for computers

Language is ambiguous and depends on context. For example, the word “bank” can refer to a financial institution or the side of a river. A phrase can also have different meanings depending on surrounding words, the speaker's intention, or the domain. NLP methods attempt to use context and patterns to reduce these difficulties.

2. Text Preprocessing

Before text is used by many NLP systems, it may be cleaned and divided into manageable units. The exact preprocessing steps depend on the task. Removing punctuation or stop words is not always appropriate because these elements can carry meaning.

Important techniques

Tokenization: Splitting text into units such as sentences, words, or subword tokens. Example: “NLP is useful” can be split into “NLP”, “is”, and “useful”.

Lowercasing: Converting text to lowercase when case does not matter. It can reduce differences between “Search” and “search”.

Stop-word handling: Deciding whether frequent words such as “the” and “is” should be retained. The decision depends on the task.

Stemming: Reducing words to a stem, sometimes producing a form that is not a standard word. For example, “connected” and “connecting” may be reduced to “connect”.

Lemmatization: Mapping a word to its dictionary base form using linguistic information. For example, “was” may be mapped to “be”.

Chunking: Dividing a long document into smaller text passages so that retrieval systems can find relevant sections.

For PDF question answering, chunks should usually retain metadata such as the source filename and page number. That metadata allows a system to show where a retrieved passage came from.

3. Text Embeddings and Semantic Search

A text embedding is a numerical vector representing a piece of text. Embedding models are trained so that useful relationships between texts can be represented in vector space. Depending on the model and training method, passages with similar meanings may have vectors that are close together.

Keyword search versus semantic search

Keyword search	Semantic search
Looks for matching words or terms.	Looks for passages with related meaning.
May miss a relevant passage when the question uses synonyms.	Can retrieve relevant passages even when wording differs.
Example: search for “car” finds the exact word.	May also find a passage about “automobiles” if the embeddings represent them as related.

A typical semantic retrieval pipeline

1. Extract text from each PDF page.
2. Split the text into overlapping chunks and attach filename/page metadata.
3. Convert each chunk into an embedding vector.
4. Store the vectors in a vector index, such as FAISS.
5. Convert the user's question into an embedding using the compatible model.
6. Retrieve the most similar chunks and inspect their relevance.
7. Use the retrieved passages to answer the question and cite the source pages.

Similarity scores

Vector search systems often compare vectors using cosine similarity, inner product, or distance measures such as Euclidean distance. A similarity score is a ranking signal, not proof that a passage fully answers the question. A system should also handle cases where no retrieved passage is relevant enough.

4. Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) combines information retrieval with a language model. Rather than relying only on information stored in a model's parameters, the system first retrieves relevant passages from a knowledge source, then uses those passages as context when preparing an answer.

Main stages in a PDF RAG application

Ingestion: Upload documents and extract their text.

Preparation: Clean the text, create chunks, and preserve page metadata.

Indexing: Create embeddings and store them in a searchable vector index.

Retrieval: Find chunks relevant to the user's question.

Answering: Answer from retrieved evidence, ideally with source citations.

Validation: Check whether the evidence supports the answer; report when the document does not contain enough information.

Grounded answers and citations improve traceability, but citations must point to passages that actually support the claim. If a PDF does not contain the requested information, a reliable application should say that it could not find enough evidence instead of inventing an answer.

5. Evaluation, Limitations, and Applications

How to evaluate a document question-answering system

Retrieval relevance: Do the top retrieved passages contain the information needed to answer the question?

Answer correctness: Does the answer match the evidence in the document?

Citation accuracy: Do the cited filenames and page numbers point to the supporting passages?

Unsupported-question handling: Does the system clearly admit when the answer is not present?

Deletion correctness: After a PDF is deleted, can its content still appear in search results or answers?

Robustness: Does the system handle empty PDFs, scanned PDFs, duplicate files, and extraction errors appropriately?

Common limitations

• Scanned PDFs may require Optical Character Recognition (OCR) if they do not contain selectable text.

• Poor chunk boundaries can separate a fact from its explanation or heading.

• Embedding models can miss domain-specific relationships or nuanced questions.

• A language model can produce a plausible but unsupported answer unless the system constrains it to the retrieved evidence.

• A citation can be formatted correctly but still be wrong if page metadata is lost or mismatched.

Applications of NLP

NLP is used in customer-support chatbots, search engines, translation tools, accessibility systems, document analysis, email filtering, voice assistants, sentiment analysis, and knowledge-management applications. In education, document question answering can help students locate explanations in lecture notes and textbooks.

6. Sample Questions for Testing DocuMind AI

1. What does NLP stand for?

2. Name four common tasks performed by NLP systems.

3. What is the difference between stemming and lemmatization?

4. Why might a PDF question-answering system split text into chunks?

5. What is a text embedding?

6. How does semantic search differ from keyword search?

7. What is the role of a vector index such as FAISS?

8. List the main stages of a RAG pipeline.

9. Why should retrieved passages preserve filenames and page numbers?

10. What should the system do if the uploaded document does not contain the answer?

11. Why is a similarity score not enough to prove that an answer is correct?

12. Name two reasons why text extraction from a PDF might fail.

13. What is OCR, and when might it be needed?

14. How can you test whether deleting a PDF really removes its content from retrieval?

15. Give two examples of real-world NLP applications.

Glossary

Term	Meaning
NLP	Natural Language Processing; methods for working with human language.
Embedding	A numerical vector representing text or another data item.
Chunk	A smaller passage created from a larger document.
Vector index	A data structure used to search vectors efficiently.
RAG	Retrieval-Augmented Generation; retrieval followed by evidence-informed generation.
OCR	Optical Character Recognition; extracting text from images of text.
Citation	A source reference that helps readers verify an answer.